

Fraud Detection in Financial Transactions: A Comparative Analysis of Classical and Neural Methods

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1. Introduction

Financial fraud detection is critical for institutions facing billions in annual losses. This report analyzes a synthetic dataset of 12,000 transactions using four approaches: linear regression (transaction amount prediction), logistic regression (fraud classification), neural networks, and autoencoder anomaly detection. All methods are implemented in a modular object-oriented pipeline. We compare supervised and unsupervised techniques, evaluating their suitability for production deployment where interpretability, accuracy, and adaptability must be balanced.

2. Data and Preprocessing

The dataset contains 12,000 transactions with 24 variables: customer demographics (age, income, tenure), transaction attributes (merchant type, channel, region, amount, amount_ratio), behavioral indicators (foreign, card present, high-risk merchant, night/weekend flags), temporal features, historical patterns (previous amounts, transaction counts), and risk scores. The target variable is binary fraud (2.13% fraud rate).

Missing values (region 0.8%, avg_prev_amount 1.2%, days_since_last_txn 1.0%) are imputed using median (numeric) or mode (categorical). **Categorical variables** (merchant_type, channel, region) are one-hot encoded. **Numeric features** are standardized using StandardScaler. Variables loss_amount and amount are excluded from fraud models to prevent target leakage.

A **stratified 70/30 train-test split** preserves the 2.1% fraud proportion in both sets (training: 8,400 obs, 179 frauds; testing: 3,600 obs, 77 frauds).

3. Descriptive Analysis

The **fraud rate of 2.13%** (256/12,000) creates severe class imbalance, making accuracy misleading—a classifier predicting all transactions as legitimate achieves 97.9% accuracy while detecting zero fraud. We therefore focus on **precision, recall, and F1-score**.

Transaction amounts are right-skewed (mean 63.0, median 54.1, range 1–730), typical of consumer spending. Fraudulent transactions show systematically higher amounts (86.7 vs 62.7), `amount_ratios` (2.31 vs 1.60), and `risk_scores` (43.9 vs 28.6), indicating fraud targets larger-than-typical transactions.

Visualizations confirm: (1) amount distribution is concentrated below 100 with long right tail; (2) fraud class is highly imbalanced; (3) grocery and restaurant dominate merchant types; (4) fraud transactions have higher median amounts with more outliers; (5) `risk_score` and `amount_ratio` correlate moderately with fraud ($r \approx 0.3\text{--}0.5$), while demographics show weak associations.

Key insight: `Risk_score` and `amount_ratio` are primary fraud drivers, justifying their inclusion in all models.

4. Linear Regression

To demonstrate baseline modeling, we estimate linear regression for **transaction amount** (continuous outcome) using all features except `amount`, `amount_ratio`, and `loss_amount`.

Performance: RMSE = 35.20, $R^2 = 0.566$ (explains 56.6% of amount variance).

Top predictors: `risk_score` ($\beta=52.98$), merchant categories (gaming, travel, fashion), `avg_prev_amount` ($\beta=12.23$), foreign/night flags. Higher risk scores and previous spending patterns drive larger transaction amounts.

Limitation: Linear regression is inappropriate for binary fraud classification—it produces unbounded predictions and violates distributional assumptions. This motivates logistic regression and neural networks for fraud detection.

5. Logistic Regression

Logistic regression models fraud probability using all features except `amount` and `loss_amount`, with `class_weight='balanced'` to address 2.1% fraud imbalance.

Performance: Test set achieves perfect classification—TN=3,523, FP=0, FN=0, TP=77; Precision=1.000, Recall=1.000, F1=1.000, AUC=1.000.

Caveat: Perfect performance reflects synthetic data where `risk_score` and `amount_ratio` linearly separate classes. Real-world systems achieve $F1 \approx 0.6\text{--}0.8$ due to evolving fraud tactics, label noise, and ambiguous cases.

Why not accuracy? A naive "always predict non-fraud" classifier achieves 97.9% accuracy while catching zero frauds. **Precision** (fraction of flags that are true fraud) and **recall** (fraction of frauds caught) are more informative in imbalanced settings.

Advantages: Logistic regression provides coefficient interpretability (regulatory compliance), fast inference (<1ms), and stable training—making it the industry baseline.

6. Object-Oriented Programming

The entire workflow is encapsulated in a `FraudAnalysisPipeline` class, providing:

Encapsulation: Data (splits, scalers, models) and operations (preprocessing, training, evaluation) bundled together

Reusability: Instantiate on new datasets without code duplication

State management: Store intermediate results for inspection

Consistent interface: All methods follow identical signatures

Class structure: `__init__` (accepts `DataFrame`), `preprocess()` (imputation, encoding, scaling, split), `descriptive_statistics()`, `fit_linear_model()`, `fit_logistic_model()`, `fit_nn_classifier()`, `fit_autoencoder()`, `evaluate_model()`.

Usage:

```
pipeline = FraudAnalysisPipeline(df)
pipeline.preprocess()
pipeline.fit_logistic_model()
pipeline.evaluate_model()
```

This design follows software engineering best practices: separation of concerns, defensive programming, and extensibility for additional models or API deployment.

7. Neural Network Classifier

A feedforward network (32 input → 64 ReLU → 32 ReLU → 1 sigmoid) provides non-linear classification alternative. Trained with binary cross-entropy, Adam optimizer (lr=0.001), class weights ($w_1=23.81$ vs $w_0=0.511$), 20 epochs, batch size 256.

Performance: TN=3,523, FP=0, FN=1, TP=76; Precision=1.000, Recall=0.987, F1=0.993, AUC=1.000. Near-perfect but marginally worse than logistic (1 false negative).

Interpretation: On this linearly separable synthetic data, neural network's non-linear capacity offers negligible advantage over logistic regression. In real data with complex feature interactions, neural networks would excel.

Trade-off: Neural networks are "black boxes" complicating regulatory compliance and customer explanations, favoring logistic regression for customer-facing decisions.

8. Autoencoder Anomaly Detection

Motivation: Supervised models require labeled fraud, but labels are delayed, noisy, and miss novel tactics. Autoencoders train only on normal transactions to learn typical patterns, flagging high reconstruction error as anomalies.

Architecture: 32 → 32 ReLU → 16 bottleneck → 32 ReLU → 32 (trained on 8,221 non-fraud transactions, MSE loss, Adam, 30 epochs). **Threshold:** 99th percentile of normal reconstruction errors (0.0903).

Performance: TN=3,480, FP=43, FN=5, TP=72; Precision=0.626, Recall=0.935, F1=0.750, FPR=1.2%.

Interpretation: Catches 93.5% of frauds without labels—remarkable for unsupervised learning. Trade-off: 43 false positives (1.2% FPR) require manual review but acceptable operationally.

Deployment: Use as first-stage screener flagging anomalies, followed by supervised models for precise classification and rule-based filters for compliance.

9. Comparison and Recommendations

Table 1: Model comparison on test set (n=3,600)

Model	Precision	Recall	F1	Notes
Linear Regression	–	–	–	RMSE 35.2, R ² 0.566
Logistic Regression	1.000	1.000	1.000	Perfect on synthetic data
Neural Network	1.000	0.987	0.993	1 false negative
Autoencoder	0.626	0.935	0.750	Unsupervised, 43 FP

Synthetic data caveat: Near-perfect supervised performance (F1≥0.99) reflects linear separability of risk_score/amount_ratio, not realistic expectations. Real systems achieve F1≈0.6–0.7 (logistic), 0.65–0.75 (neural), with autoencoders at precision 0.3–0.5 and recall 0.7–0.85.

Production recommendation: Layered system combining (1) autoencoder anomaly screening (catches novel fraud), (2) logistic regression baseline (interpretable, regulatory-compliant), (3) neural network refinement (for complex cases), (4) rule-based overrides (explainability).

Key insight: Risk_score and amount_ratio dominate predictive power—invest in maintaining risk models and real-time behavioral profiling.

Limitations: Synthetic data, no temporal dynamics, limited customer features. Future work: ensemble methods, graph neural networks for fraud rings, cost-sensitive learning, online model updates, explainability tools (SHAP/LIME).

10. Conclusion

This analysis demonstrates fraud detection using classical statistics (linear/logistic regression), neural networks, and unsupervised autoencoders. Supervised models achieve F1≥0.99 on synthetic data, with logistic regression providing superior interpretability. Autoencoders achieve 93.5% recall without labels, valuable for novel fraud. The FraudAnalysisPipeline class provides production-ready modular implementation. Real-world deployment requires layered

architectures balancing accuracy, interpretability, and adaptability, with continuous retraining and ethical oversight for fairness and privacy compliance.